Keaton Lees

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Experience

Systems and Controls Engineer
Ballard Power Systems | Burnaby, BC *Aug 2020 - Dec 2020*
- Worked with a team of engineers to develop and improve control systems for hydrogen fuel cells
- Assembled and tested components for use in prototype fuel cells
- Programmed Python scripts to update and maintain sensor database

Systems Engineer
Ballard Power Systems | Burnaby, BC *Jan 2020 - Apr 2020*
- Assisted in developing and testing control systems for hydrogen fuel cells using Arduino and RaspberryPi
- Sourced components worth over $100,000 for the department

Founder
kleez3DPrints | Surrey, BC *Summer 2019*
- Utilized SolidWorks and UltimakerCura to deliver >$500 worth of 3D-printed designs including figurines, team logos, and puzzles

Competitions

Hack the North 2020++
Hackathon | Our project: ToDoIt *Jan 2021*
- Programmed ToDoIt, a time management tool that predicts the time required to complete tasks using Flask, React and Python

Amazon + Adobe Creative Jam
Designathon | Our project: Rethink *Oct 2020*
- Prototyped Rethink, an application that introduces high school students to design, using Adobe XD

TOHacks 2020 Hackathon
Hackathon | Our project: Puppr *May 2020*
- Developed Puppr, a digital dog adoption platform using HTML, CSS, Javascript and live web-scraping

Projects

Smart Home Clock
Flask, Python, HTML, CSS, Raspberry Pi *Nov 2020*
- Designed and programmed a Smart Clock using Flask, Raspberry Pi and a touchscreen display
- Added the ability to fluidly control and interact with smart home functions using APIs

Donkey Kong Re-Creation
Java, HTML *2019*
- Re-created the 80s Donkey Kong arcade game using Java and JGEngine and integrated external gyroscopic controls

Education

Systems Design Engineering 2024
University of Waterloo
Relevant Courses: Digital Computation; Data Structures and Algorithms; Digital Circuits; Human Factors in Design; Design, Systems and Society; Engineering Prototyping

Skills

Languages
C++
Java
Python
Javascript
HTML
CSS
React
Arduino C

Tools
Flask
Bootstrap
Git
Raspberry Pi
Arduino
3D-printing
SolidWorks
Cura

Achievements

Top 20 - Amazon Creative Jam
Top 20 out of 420 teams for our project, Rethink

Nominee - Schulich Leaders
Sole School representative for the Schulich Leaders Scholarship

Medalist - BC Summer Games
Gold and Bronze medalist in Track and Field at the BC Summer Games

3rd Degree Black Belt
3rd Degree Black Belt in Taekwondo

Interests

- 3D-printing
- Arduino and Raspberry Pi
- Track and Field - Athlete and Official
- Indoor and Beach Volleyball

Jan 2020